

Catherine Martlin

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EDUCATION

SWARTHMORE COLLEGE

BA IN ASTRONOMY AND CLASSICAL STUDIES | GRAD. 2015

INTERCOLLEGIATE CENTER FOR CLASSICAL STUDIES

Study Abroad Fall 2014 | Rome, Italy

SKILLS

Languages/Tools: Python, Git, Jupyter, WebApp Development (Flask, Django), Pandas, NumPy, Astropy, Astroquery, LATEX, Bash, IDL, FORTRAN, IRAF, SQL, CI/CD (GitHub Actions), Agile/Scrum, Containerized Environments(Docker), AWS Batch • **Documentation:** ReadTheDocs, HDoc, Confluence, Sphinx • **Other:** FITS files, data file pipelines, Image calibration, PSF modeling, Quicklook systems/Databases, Drizzlepac, CRDS(Reference File System)

WORK EXPERIENCE

STAFF SCIENTIST II, SCIENCE SUPPORT | SPACE TELESCOPE SCIENCE INSTITUTE

June 2015 - Current Date. | Baltimore, MD

Core Responsibilities

- Developed, deployed, and maintained high-throughput image calibration pipelines for HST's WFC3 instrument using various tools such as Python, Jupyter Notebooks, Git and SQL.
- Contributed to internal team Quicklook systems, automated monitoring, anomaly detection, and on-call operations for daily image quality control.
- Led translation of legacy IDL and FORTRAN image processing algorithms into modern Python pipelines with CI/CD integration through Github.
- Regularly interfaced with cloud-style environments and remote data systems for reference file validation and delivery.
- Wrote clear technical documentation and tutorials, available internally and externally as Jupyter Notebooks, to ensure knowledge is easily shared and processes stay consistent over time.
- Collaborated with multiple teams of engineers, operators, and scientists to test and approve pipeline updates, ensuring they met performance and quality standards.

Project Highlights

- WFC3 Quicklook System: Key contributor and former team lead—built new Python monitors, managed GitHub workflows, and oversaw SQL database health.
- Geometric Distortion Pipeline: Modernized distortion solution generation using Python; delivered 15+ geometric distortion calibration files (IDCTAB) across multiple filters— which is core to precision astrometry.
- Postflash Calibration: Annual delivery of updated calibration reference files using automated notebooks and image QA routines previously written by me.
- DASH Pipeline: Developed alignment system for non-guided HST exposures; released as public-facing Jupyter notebook tutorial and Instrument Science Report.

Broader Contributions:

- Mentored interns and newer team members, helping them grow into key contributors for operations and support tasks. Following a 9-month internship, intern was hired full-time by STScI.
- Organized and taught technical workshops (through the international organization The Carpentries) to improve coding and data skills of colleagues across the institute.
- Contributed to JWST Quicklook support (2018–2019), showing adaptability to different telescope systems and needs.

RESIDENT ASSISTANT SELECTION COMMITTEE; CLERY COMPLIANCE COMMITTEE |

SWARTHMORE

Spring 2015 | Swarthmore College

- 1 of 6 selected to serve on RA selection Committee and 1 of 2 selected to serve on Clery Compliance Committee.
- Interviewed potential RAs; reviewed applications; selected the next year's student leadership.
- Met with deans and public safety to review the college's bylaws and reports of sexual assault to ensure compliance with the Clery Act.

STUDENT BUDGET COMMITTEE MEMBER | SWARTHMORE COLLEGE

Fall 2013 - Spring 2014 | Swarthmore College

- Selected as part of 8 person committee in charge of handling over \$500,000 in student activities fees.
- Lead Spring Budgeting where all students groups submit budgets and are given appropriate funding for the next year.
- Met weekly to allocate supplemental funds to student groups throughout the year.

RESIDENT ASSISTANT | SWARTHMORE COLLEGE

Fall 2013 - Spring 2014 | Mary Lyon Dormitory

- Leader of training groups and sessions during Freshman Orientation and of weekly 'study breaks' through the year.
- Responsible for helping over 30 students on my hall with day to day residential issues as well as more unusual and intense issues and with contacting administration.

TELESCOPE OPERATOR | SWARTHMORE COLLEGE

Fall 2011 - Spring 2013 | Swarthmore Astronomy

- Helped set up the dome, telescope, cameras and software for taking data for department research in the evening.
- Ensured nothing went wrong during the several hours of time spent taking data. (e.g. telescope loses guide star, becomes too cloudy out for usable data, etc.)

HEAD GRADER | SWARTHMORE PHYSICS AND ASTRONOMY

Fall 2013 - Spring 2014; Spring 2015 | Swarthmore College

- Organized other student graders and graded homework for 30+ students for 3 semesters of introductory Astronomy/Physics courses.
- Briefed Professor regarding homework questions that needed more clarification based on overall class understanding and on progress of individual students.

CONGRESSIONAL PAGE | U.S. HOUSE OF REPRESENTATIVES

Fall 2009 - Spring 2010 | Washington, D.C.

- Answered questions about votes on the House floor; which member was currently speaking on the floor.
- Delivered messages to members on the floor. Delivered legislation from members' offices to the House floor.

RESEARCH EXPERIENCE

WESLEYAN ASTRONOMY - X-RAY/OPTICAL IMAGING PIPELINE DEVELOPER Summer 2014 | Middletown, CT

Wesleyan University – Created a pipeline for matching Chandra X-ray sources to HST counterparts and deriving spectral parameters for 4500+ source epochs. Tools used: CIAO, Python, FITS, cross-catalog matching, astrophysical data modeling.

SWARTHMORE ASTRONOMY - M-DWARF ROTATIONAL STUDY LEAD Summer 2013; Spring 2014 | Swarthmore, PA

Led a spectroscopic data analysis project involving over 150 young stars; tools included IDL, IRAF, Python.

AWARDS

2024 Space Telescope Science Institute **Achievement Award: DEI Team Award: Neurodiversity Starter Group**
2021 Space Telescope Science Institute **Bravo Award: Promotion to Staff Scientist, Science Support II**
2021 Association of Universities for Research in Astronomy **Achievement Award: JWST Cycle 1 User Support Team**
2020 Space Telescope Science Institute **Achievement Award: DEI Team Award: DCRWG**
2020 Space Telescope Science Institute **Bravo Award: ExoCTK Version 1.1 Release**
2020 Space Telescope Science Institute **Bravo Award: WFC3 Photometry Working Group Award**
2019 Space Telescope Science Institute **Achievement Award: Successful Deployment of ExoCTK**
2018 Space Telescope Science Institute **Bravo Award: Completing the Application Security Class**
2016 Space Telescope Science Institute **Bravo Award: Promotion to Research and Instrument Analyst II**
2011 National **Questbridge Scholar** - Matched with Swarthmore College
2010 US House of Rep. Page School Jefferson Scholar Award

ACTIVITIES AND SOCIETIES

2022 to Current date Open Water Diving Certification, Enriched Air Diver Training (PADI)
2018 to Current Date Bocce Ball Team Member run through Community Charity Organization Volo
2013 to 2015 Sigma Xi Research Society
Fall 2011 to Spring 2015 Swarthmore Fencing Team *Captain (2012 - 2014) & Co-President (Spring 2014)*
2011 Dreamfar High School Marathon Team
2007 to Current Date American Legion Auxiliary